INCH-POUND
MIL-C-60537A(AR)
17 November 1993
SUPERSEDING
MIL-C-60537(MU)
23 August 1966

MILITARY SPECIFICATION

Cartridge, Grenade: 5.56MM, M195

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for the use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. Scope. This specification covers the requirements examinations and test for Cartridge, Grenade: 5.56MM, M195 intended for propelling grenades from 5.56 mm rifles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1330

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-414	-	Sampling Procedures and Tables for Inspections by Variables for Percent Defective
MIL-STD-636	-	Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	-	Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

U.S. ARMY ARMAMENTS RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

DRAWINGS

Munitions Command

7643674	-	Classification of Cartridge Case Defects
10534926	-	Cartridge, Grenade: 5.56MM, M195
12551963	-	Packing and Marking for Box, Wirebound, for Cartridge, 5.56MM
IEL-10534926	-	Inspection Equipment List for Cartridge, Grenade, 5.56MM, M195

PUBLICATIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

SCATP-5.56mm - Test Procedures for 5.56mm Cartridges

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 General. The cartridge shall comply with drawing 10534926, referenced specifications, and the requirements stated in this section.

3.2 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.3 Residual stress. The sample cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Waterproof. Cartridges shall have a minimum velocity of 140 fps after being submerged under twelve (12) inches of water for 24 hours.

3.5 Velocity. The sample cartridges conditioned for a minimum of 4 hours and fired in a horizontal position at the indicated temperatures shall propel grenades weighing 1.56 pounds plus or minus 0.01 pounds (1.56 ± .01) at velocities specified below. The velocity shall be measured at a point 5 feet, 6 inches beyond the forward end of the grenade when fully positioned for launching.

Degree Fahrenheit (F)	Instrumental Velocity Ft per Second (ft/sec) Minimum Individual	Maximum Individual
65° to 75°	140	165

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

125° to 130°	140	165
-40° to -45°	140	165

3.6 Function and Casualty. The sample cartridges shall function without casualty at temperatures of minus 40 degrees to minus 45 degrees F., at ambient temperature, and at plus 125 degrees to plus 130 degrees F.

3.7 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and in 3.7.1 and 3.7.2.

3.7.1 Metal defects. The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.7.2 Foreign matter. The cartridge shall be free of corrosion, stains, discoloration, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Part Description	Drawing	Quantity
Cartridge, Grenade: 5.56MM, M195	10534926	550

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens shall be subjected to the examinations and tests listed in Table I and may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

TABLE I First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-60537A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Cartridge, Grenade: 5.56MM, M195	1 OF 1	10534926
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Examination of Defects			
	Critical	550, 0-1	3.1	4.4.2.1
	Major	50, 0-1	3.1	4.4.2.1
	Minor	28, 0-1	3.1	4.4.2.1
	Tests (see 4.4.3)		3.3-3.6	4.5.1-4.5.5

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lot formation shall comply with MIL-A-48078.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels	
	I	II
91 to 150	50	19
151 to 280	50	19
281 to 500	50	22
501 to 1,200	50	28
1,201 to 3,200	119	31
3,201 to 10,000	119	39
10,001 to 35,000	136	47
35,001 to 150,000	172	57
150,001 to 500,000	198	65
500,001 and over	244	65

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.9). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, Grenade: 5.56MM, M195	1 OF 2	10534926
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST 1/	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Split case	100%	3.1	Gage(SCAMP Line)/ Visual(Conventional)
2	Perforated Case	100%	3.1	Gage(SCAMP Line)/ Visual(Conventional)
3	Missing Wad	100%	3.1	Visual
4	Weight, min	100%	3.1	Weighing
Special				
a	No primer 2/, 3/	100%	3.1	Visual
b	Cocked primer 2/, 3/	100%	3.1	Visual
c	Inverted primer 2/, 3/	100%	3.1	Visual
Major				
101	Corroded or stained, if etched	Level I	3.1	Visual
102	Round head	Level I	3.1	Visual
103	Beveled underside of head	Level I	3.1	Visual
104	No chamfer on head (rim)	Level I	3.1	Visual
105	Loose primer	Level I	3.1	Visual
106	Wad lacquer missing or inadequate	Level I	3.1	Visual
107	Opening between wad and case wall	Level I	3.1	Visual
108	Cartridge profile failure (requiring more than 20 pounds dead weight to insert in profile and alignment gage)	Level I	3.1	8654100,8654102 8654108,8654109 8654110,8654111 8654112

NOTES:

1/ Refer to MIL-STD-636, defects 1 through 38, for visual standards. While no photographs of grenade cartridges are included in this Standard, the cartridge case defects illustrated in the Nato Caliber 7.62mm section can be used. In the event of conflict between this specification and MIL-STD-636 as to defect classification, the classification specified in this specification shall apply.

2/ Special defects are defined in paragraph 6.5.4 of MIL-A-48078.

3/ The visual defect standards in MIL-STD-636 shall be used except "critical" shall be changed to "special".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-C-60537A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge, Grenade: 5.56MM, M195	2 OF 2	10534926
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
109	Thickness of head, min	Level I	3.1	8654104
110	Depth of primer	Level I	3.1	8654105
111	Case dent	Level I	3.1	Visual
112	Draw Scratch	Level I	3.1	Visual
113	Scaly metal	Level I	3.1	Visual
Minor				
201	Cartridge discolored, dirty, oily	Level II	3.1	Visual
202	Case dent	Level II	3.1	Visual
203	Draw scratch	Level II	3.1	Visual
204	Scratch	Level II	3.1	Visual
205	Scaly metal	Level II	3.1	Visual
206	Fold	Level II	3.1	Visual
207	Wrinkle	Level II	3.1	Visual
208	Bulge	Level II	3.1	Visual
209	Illegible or missing head stamp	Level II	3.1	Visual
210	Defective head	Level II	3.1	Visual
211	Nicked or dented primer	Level II	3.1	Visual
212	No waterproofing material (primer pocket joint)	Level II	3.1	Visual
213	Defective primer crimp	Level II	3.1	Visual
214	Mouth crimp missing or incorrect	Level II	3.1	Visual
215	Diameter of extractor groove	Level II	3.1	8654102
216	Diameter of head	Level II	3.1	8654103
217	Thickness of head, max.	Level II	3.1	8654104
218	Workmanship	Level II	3.1	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE TESTING

CLASSIFICATION OF CHARACTERISTICS

MIL-C-60537A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Cartridge, Grenade: 5.56MM, M195	1 OF 1	10534926
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Tests (see 4.4.3)		3.3-3.6	4.5.1-4.5.5

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.3 Tests. The tests listed in Table III shall be conducted in accordance with the methods and procedures specified in 4.5.

4.4.3.1 Test samples. The quantities for the various tests shall be as specified in Table III. Test cartridges shall be selected in such a manner that the sample is representative of the entire lot. Only critical defectives shall be removed. The cartridges shall be thoroughly mixed before being divided into the samples for the various tests.

TABLE III. Number of Cartridges

Test	Temperature °F. Ambient Temp.	65 to 75	-40 to -45	125 to 130	Requirement Paragraph
Residual Stress (Mercurous Nitrate) 1/	50	--	--	--	3.3
Waterproof 2/	50	--	--	--	3.4
Velocity	--	25 3/	25 3/	25 3/	3.5
Function and Casualty 4/					3.6

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. If any failing cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of ten or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If more than three, but less than ten cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, ten or more cartridges fail to comply with the applicable requirement.

3/ The variables sampling plan of MIL-STD-414, Table C-3, for an M value of .827 for the lower limit and an M value of 1.95 for the upper limit (see example C-4 for method of calculation).

4/ The lot shall be rejected when function and casualty defects noted in the firing of the velocity test exceed the acceptance

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

number for the cumulative sample in Table IV. If the number of defects found in the initial sample exceeds the acceptance number for the initial sample of Table IV but is equal to or less than the acceptance number for the cumulative sample (Table IV), a second sample consisting of double the quantity specified for the velocity test, shall be fired in the weapon specified therefore. If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If in testing the second sample, defects other than those for which the second sample is being tested, should occur to the extent that they exceed the acceptance numbers for the cumulative sample, the lot shall be rejected.

4.4.3.2 Firing defects. Firing defects and acceptance numbers shall be as specified in Table IV.

TABLE IV. Firing defects

Defects	Acceptance Number	
	First Sample	Cumulative First and Second Sample
1. Misfire	0	1
2. Primer Leak		
a. Perforation in firing pin indent in primer cup	1	5
b. Escape of gas through primer cup other than (a)	0	2
c. Escape of gas around primer cup 1/	1	2
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	1	2
3. Case Casualties		
a. Longitudinal split 3/		
(1) Body (J)	1	2
(2) Body (K)	0	2
(3) To head (L)	0	1
(4) Through head (M)	0	1
b. Circumferential rupture 3/		
(1) Partial, body (J)	0	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

(4) Complete	0	1
4. Failure to extract	0	1

1/ Gas escapes around more than 50 percent of periphery of cup.

2/ No penalty shall be charged against a fired case with mouth partially or completely burned. If, however, such fired cartridge case shows the occurrence of some other firing casualty, exclusive of failure of rosebud crimp to open, such other casualty shall be penalized in accordance with Table IV.

3/ For location of defects indicated by letters in parentheses, see Drawing 7643674.

4.4.4 Packaging, packing and marking inspection. During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a dummy, blank or any bullet type cartridge shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.4.5 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Residual stress (Mercurous Nitrate). The test shall be conducted in accordance with SCATP-5.56MM.

4.5.2 Waterproof. The water in the tank shall be between 65°F and 75°F for the duration of the test. The cartridges shall be submerged upright to a depth of 12" from the top of the cartridge for 24 hours. After 24 hours, the cartridges shall be removed from the water bath and towel dried. The cartridges shall then be fired for velocity in accordance with Appendix A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.5.3 Velocity. The test shall be conducted in accordance with Appendix A.

4.5.4 Function and casualty. Observation for firing defects listed in Table IV shall be made during the firing of the velocity test. Retest shall be fired in accordance with the procedure prescribed for the velocity test in Appendix A, except that all reference pertaining to obtaining of velocity data shall be disregarded.

4.5.5 Temperature tests. The test shall be conducted in accordance with Appendix B.

4.5.6 Defect penalty. If, in the velocity test, the occurrence of a firing defect prevents the obtaining of a reliable result, an additional shot shall be fired. The velocity test shall not be penalized, but the acceptance sample shall be penalized for such defects in accordance with Table IV.

5. PACKAGING

5.1 Packing, level A. (Worldwide shipment) The cartridges shall be packed in accordance with drawing 12551963.

5.2 Marking and labeling. Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The components covered by this specification are intended for use in the Cartridge, Grenade: 5.56MM, M195.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

c. Requirements for submission of first article sample.

d. Type and level of packing (see 5.1).

e. Provision for submission of Inspection Equipment Designs.

f. Provision for submission of detailed inspection plan for the cartridge (see 6.7).

g. Provisions for submission of acceptance inspection results for each lot of ammunition presented to the Government (see 6.6).

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Hazard notice. The cartridge described herein and some of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measure to prevent fire, explosion, adverse environment, rough handling, corrosive atmosphere, or electrically induced incidents. Such measures shall include the employment of an effective safety program that addresses the inherent hazards associated with the cartridge.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal and Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.6 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the U.S. Army, one (1) copy of all ballistic data and the ammunition data card for each lot should be forwarded to: Commander, ARDEC, ATTN: SMCAR-QAF-S, Picatinny Arsenal, NJ 07806-5000.

6.7 Submission of detailed inspection plan for review. The detailed inspection plan for the cartridge should be submitted to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

the responsible U.S. Government quality assurance element.

6.8 Subject term (key word) listing.

Ammunition
Blank
Small Arms
Testing

6.9 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.10 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 1330-A378)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

APPENDIX A

Velocity, Function and Casualty Tests

10.0 SCOPE

10.1 Scope. This appendix gives the test procedure for performing the velocity, function and casualty test procedure tests.

20.0 APPLICABLE DOCUMENTS
This section is not applicable to this appendix.

30.0 PURPOSE

30.1 Purpose. The purpose of the tests is the following:

a. The purpose of the velocity test is to determine if the velocity of rifle-grenades, when propelled by the grenade cartridges, complies with the velocity requirement contained in Section 3.

b. The purpose of the Function and Casualty test is to demonstrate, by firing in weapons of representative type, whether or not the ammunition undergoing acceptance can be expected to perform satisfactorily in the service weapon for which it has been designed.

c. Casualties and malfunctions can be caused either by the ammunition or by the weapon in which it is fired, so that, to a certain extent, these two factors are inter-dependent. A faulty or poorly adjusted weapon can cause casualties in normal ammunition, but if the weapon be in proper condition when casualties are encountered, then the fault lies with the ammunition.

d. The velocity and function and casualty tests are fired simultaneously.

40.0 EQUIPMENT

40.1 Equipment. Equipment listed in the Velocity section of the appropriate Inspection Equipment List shall be used.

40.2 Firing range. The firing range shall be arranged as shown on Chart #1, at the end of this section. The maximum distance between the lumiline screens shall be ten feet.

40.3 Pier or mount Pier or mount shall be of solid construction on which the test fixture assembly is mounted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

APPENDIX A

50.0 TEST PROCEDURE

50.1 Pre-firing (Preparative for test). The following test procedure shall be followed:

a. Weapons shall be of the latest design or most recent issue. The use of weapons or parts of earlier design is sometimes permitted for special purposes.

b. Prior to assembling the velocity test weapon in the test fixture, the applicable magazine (empty) shall be inserted in the weapon; bolt shall be in open position and the safety in "safe" position.

c. The test weapon shall be assembled in the test fixture on the mount and sighted in on the recovery container down-range. Weapon shall be sighted-in for each cartridge to be fired.

d. Test cartridges shall be selected insuch a manner that the sample is representative of the entire lot. The test cartridges shall be examined for defects. Only cartriges with critical defects shall be removed. The defects shall be noted on the test sheet form.

e. The controlled temperature containers shall be brought to the specified temperatures before the test cartridges are placed in their respective containers. The test cartridges shall be brought to a temperature of 60°F to 80°F prior to being placed in their respective controlled temperature containers.

f. The required number of test cartridges shall be placed in a vertical position, primer-end down, in recessed holding blocks. The cartridges shall be placed in the controlled-temperature containers to such a manner that all cartridges are subjected to the specified temperature for the length of time prescribed in the item specification. Cartridges to be fired at 70°F ± 5°F shall be stored in a container or room maintained at a temperature of 70°F ± 2°F with a relative humidity of 60 ± 5 percent for the time specified. The container or room shall be of sufficient capacity to allow free circulation of air.

g. Lumiline screens are checked for position. It is of the utmost importance that the lumiline screens be placed in their proper positions. Measurements must be accurate. The lumiline screen nearest the muzzle of the velocity weapon shall be of blast proof construction. Range shall be equipped with a suitable recovery container filled with cotton waste or equivalent, for stopping and preserving a propelled proof-slug, so that recovery can be made. The recovery container should be located approximately fifteen feet from the muzzle of the velocity weapon, in the line of fire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

50.2 During firing. The following test procedure shall be followed:

a. When the test is being conducted, the chamber of the weapon shall be empty, the bolt in the open position and the safety selector placed in the "safe" position, until ready to fire. This precaution is necessary because the technician is required to place himself between the muzzle of the weapon and the recovery container during certain operations.

b. The proof-slug(s) is/are placed at a point convenient to the technician. Three warming (fouling) shots having components of the same type as the test cartridges are fired and velocity readings of the proof-slugs are recorded by the chronographer to assure that the equipment is functioning properly.

c. The recessed holding block containing the test cartridges (reference cartridges are not fired) is removed from the controlled temperature container or room and placed at a point convenient to the technician, provided the temperature of the firing room is 70°F ± 5°F, otherwise the test cartridges shall be placed in an insulated box which has been conditioned at 70°F ± 5°F, and the box placed at a point convenient to the technician. Cartridges are then removed singly from the insulated box and fired. If an insulated box is not available then the cartridges shall be removed singly from the controlled temperature container or room and fired.

d. A proof-slug is fully seated on the flash suppressor on the muzzle of the weapon.

e. In order to simulate actual combat conditions the cartridges are not rotated prior to insertion into the chamber of the test weapon; however, the manner of handling the ammunition from cartridge to cartridge shall be reasonably uniform.

f. As an extra safety precaution the charging handle shall be pulled fully to the rear and held in this position, using the left hand. Using the right hand, insert the cartridge into the chamber of the weapon. Release changing handle when right hand is clear of weapon.

g. The bolt is then released by depressing the upper portion of the bolt catch with the thumb of the left hand allowing the action to close. Do not "ride" the charging handle forward with the right hand. If the charging handle is eased forward from the open position, the bolt may fail to lock and the rifle will not fire (When using the M16A1 rifle it is sometimes

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

necessary to hit the "forward assist" with the right hand in order to accomplish full closure of the bolt).

h. The lanyard is carefully attached to the trigger. If any delay should occur after the cartridge has been chambered, and the duration of the delay is such that the temperature of the cartridge has changed significantly, that cartridge shall be extracted and another inserted in it place.

i. The technician retires to a safe position and pulls the lanyard with a smooth firm motion. The velocity of the shot shall be recorded by the chronographer. The fired case is visually examined by the technician for possible case casualties.

j. The technician then recovers the fired proof-slug from the recovery container and examines it carefully. If the proof-slug is not damaged in any way, it shall be re-seated on the flash suppressor on the muzzle of the weapon and used again. Should the proof-slug be damaged in any way, it shall be replaced with a new proof-slug.

k. The procedure prescribed in f through j is followed for each cartridge to be fired.

l. Continuous air cooling shall be used on the barrel. If air cooling is not available, barrel shall be allowed to cool to ambient temperature between each series of tests, or after a maximum number of twenty-five cartridges has been fired.

m. If any of the personnel conducting the test observes any abnormality that may invalidate the velocity measurements, the circumstances shall be reported immediately through appropriate supervisory channels and the test suspended until instructions are received from proper authority. If any result appears abnormal, even though the possible cause cannot be immediately assigned, the circumstances should be reported immediately for investigation and decision by proper supervisory authority.

60.0 RECORDING OF DATA

60.1 Velocity. Individual velocities, average velocity, extreme variation and standard deviation of the individual velocities shall be recorded.

60.2 Number and type of case defects. Casualties shall be reported in accordance with terminology specified in 80.0 and the applicable specification.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:21Z
Check the source to verify that this is the currentversion before use.

60.3 Failures. Failures of gun parts shall be shown on the ammunition report and gun card.

60.4 Weapon data. The following test weapon data shall be recorded on the test sheet form:

a. Weapon number and type
b. Total number of cartridges fired in rifle.

70.0 OPERATIONAL NOTES

70.1 Malfunctions. In the event any malfunction occurs during firing of the test, a detailed check shall be made to determine whether the ammunition or the equipment is at fault. The check of the weapon shall include measurement of the firing-pin protrusion and firing-pin indent. To assist in determining whether ammunition or equipment is responsible for a malfunction, it is good practice to test the weapon in question using ammunition of known characteristics, and to test the ammunition in question by firing in another weapon of the same type. If it is established that some faulty condition of the weapon is responsible for the malfunction, then the test shall be disregarded, the weapon shall be corrected or replaced, the test shall be refired.

70.2 Misfire. If a misfire is encountered, the weapon is examined carefully to determine if the cause is attributable to the gun. In any ballistic acceptance test where a misfire occurs, a second attempt to fire a primer is not made. It is mandatory that a period of at least five minutes elapse after the misfire occurs before the action of the weapon is opened, whereupon the misfired cartridge is carefully removed in accordance with existing safety regulations, and preserved for further examination. All handling and examination of misfired cartridges shall be conducted with due regard for the hazards involved. The weapon in which a misfire occurs shall be thoroughly checked; it shall be disassembled and all component parts critically scrutinized. Results of such examination shall be included on the test report as a matter of information.

70.3 Misfired cartridges. Laboratory examination of the misfired cartridges shall be made to determine the specific

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

cause; the result of the investigation shall be included on the est report.

70.4 Cartridge cases. Upon completion of firing, all cartridge cases from the test ammunition shall be carefully examined for firing defects. If any defect is found, a detailed check of the equipment shall be made to determine whether the ammunition or the equipment is at fault. If it is established that a faulty weapon is responsible for the firing defect, then the test shall be disregarded, the weapon shall be corrected or replaced, and the tests shall be refired. If it cannot be established that the weapon or other equipment is at fault, then the firing defects shall be charged against the ammunition.

80.0 DEFINITIONS

80.1 Misfire. Failure of a round of ammunition to fire after initiating action is taken. There are two general categories of misfires:

a. The primer fails to fire when struck by the firing pin.
b. The propellant fails to ignite when the primer fires normally.

80.2 Perforated primer. A perforated primer is one in which the indent in the primer cup, made by the firing pin, is entirely perforated. It can be identified by a visible hole through the primer, or, if the perforation be minute, by discoloration of the indent caused by gas burning.

80.3 Primer leak. Discoloration caused by gas leakage around the junction between the primer cup and the primer pocket wall.

80.4 Loose primer. Looseness, but not so as to permit the fired primer to fall from the primer pocket after the cartridge is fired.

80.5 Blown primer, or a primer which falls out of the primer pocket. A blown primer is a primer which, when the cartridge is fired, is separated completely from the head of the cartridge case, and both the head of the case and the pocket are enlarged and deformed. A primer which falls out of the pocket is in the same category as a blown primer but the distortion to the primer pocket is less obvious.

80.6 Ruptured case. A circumferential separation of the case wall produced by firing. Ruptures are divided into two

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

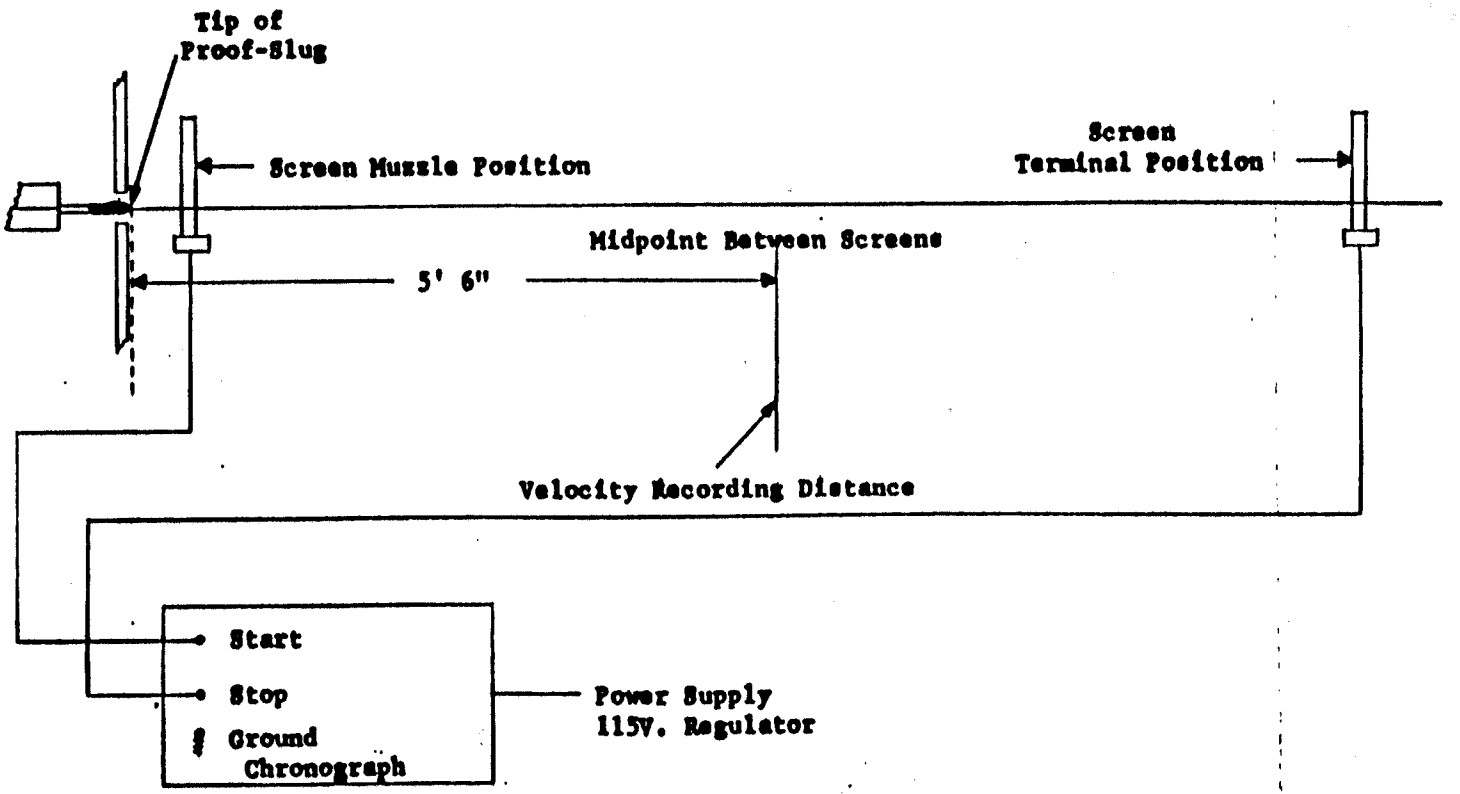
categories, partial and complete. A partial rupture is one which extends less than 360° around the case; a complete rupture is one which extends entirely around the case, separating the case into two parts. Ruptures are designated according to position, as indicated on drawing 7643674.

80.7 Split case. A longitudinal separation of the metal in the case wall, produced by firing. Splits shall be classified as prescribed by the cartridge specification.

80.8 Failure to extract. The fired case is not removed from he weapon chamber by normal gun action.

80.9 Primer setback. The backward movement of primer cup in a cartridge case which occurs when the base of the cup is not properly supported by the bolt face or breech block, and protrudes above the head of the cartridge case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.



RANGE EQUIPMENT, VELOCITY TEST FOR CARTRIDGE GRENADE, RIFLE, 5.56 MM

Chart #1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

APPENDIX A

Engineering Proof Test Record
Velocity and Function & Casualty Tests

Propellant: Ammunition:

Type
Army Lot
Charge
Case
Primer

Lot
Type
Caliber

Spec/Auth

Receiver No.	Barrel No.	Times Fired	Firing Pin Protrusion	Firing Pin Indent	Headspace
	Stored at		Stored at	Stored at	Total All Conditions
	Fired at		Fired at	Fired at	

Shot No.	Velocity	F&C*	Velocity	F&C*	Velocity	F&C*	F&C*
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							
16							
17							
18							
19							
20							
21							
23							
24							
25							
Total							
Mean							
Ex. Var.							
Std. Dev.							

* For legend, see Item Specification, no entry indicates "No Defects".

REMARKS:

Proof Technician(s)
Date Fired

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

APPENDIX B

Temperature (High & Low) Test Procedure

10.0 SCOPE

10.1 Scope. This appendix gives the test procedure for performing the high and low temperature test procedure.

20.0 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30.0 PURPOSE

30.1 Purpose. The purpose is to determine the effect of high or low temperature storage upon the ballistic performance of ammunition or components thereof.

40.0 EQUIPMENT

40.1 Equipment. Equipment listed in the temperature test section of the Inspection Equipment List shall be used.

40.2 Container. Controlled-temperature container or containers capable of maintaining temperatures within a tolerance of ± 2° F for temperature of 70°F. (Insulated boxes which have been brought to the required temperature may be used to transfer ammunition from controlled-temperature storage to the test weapons.)

40.3 Container capacity. The controlled-temperature container(s) shall be of sufficient capacity to permit conditioning the ammunition to the desired temperature, and to allow free circulation of air.

50.0 TEST PROCEDURE

50.1 Preparation for test. The same procedure used in preparation for firing Velocity, Function and Casualty (Appendix "A") shall be used for this test with the following exceptions:

a. Low temperature - The controlled-temperature container shall be cooled to the specified temperature before the test cartridges are placed inside. When the controlled-temperature container indicates the temperature specified, the ammunition shall be placed in the container in such a manner that all cartridges are subject to a uniform temperature and held thereat for the time specified in the item specification. The weapon in which the ammunition is fired shall be at ambient temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

b. High temperature - The controlled-temperature container shall be heated to the specified temperature before the test cartridges are placed inside. When the controlled-temperature container indicates the temperature specified, the ammunition shall be placed in the container in such a manner that all cartridges are subjected to a uniform temperature and held thereat for the time specified in the item specification. The weapon in which the ammunition is fired shall be at ambient temperature.

c. Temperature of test ammunition shall be 60°F to 80°F. prior to conditioning at high or low temperature.

50.2 Firing the test. The following procedure shall be followed:

a. The ammunition under test, stored at the temperature specified in the applicable specification, is removed from the controlled-temperature container and placed in the insulated box (referenced in 40.2), which has also been conditioned to the temperature specified. The insulated box is then placed at a point convenient to the technician.

b. The procedure used for firing the Velocity, Function and Casualty Test (Appendix "A") shall be used for this test; however, five cartridges are removed from the controlled-temperature container and placed in the insulated box. The cartridges are then removed singly from the insulated box and fired.

60.0 RECORDING OF DATA

60.1 Results. Results shall be recorded directly on the test report form.

60.2 Velocity. Individual velocities and the average velocity, extreme variation, standard deviation of the individual velocities, at each specified temperature.

60.3 Function and casualty. Number and type of case casualties and weapon malfunction at each specified temperature.

60.4 Temperature. Temperature at which test cartridges were stored and fired shall be recorded.

60.5 Weapon data. The following test weapon data shall be recorded on the test sheet:

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:21Z
Check the source to verify that this is the currentversion before use.

APPENDIX B

a. Weapon number and type.
b. Total number of cartridges fired in weapon prior to test.
c. Headspace measurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-60537A (AR)	931117

3. DOCUMENT TITLE: CARTRIDGE, GRENADE: 5.56MM, M195

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.